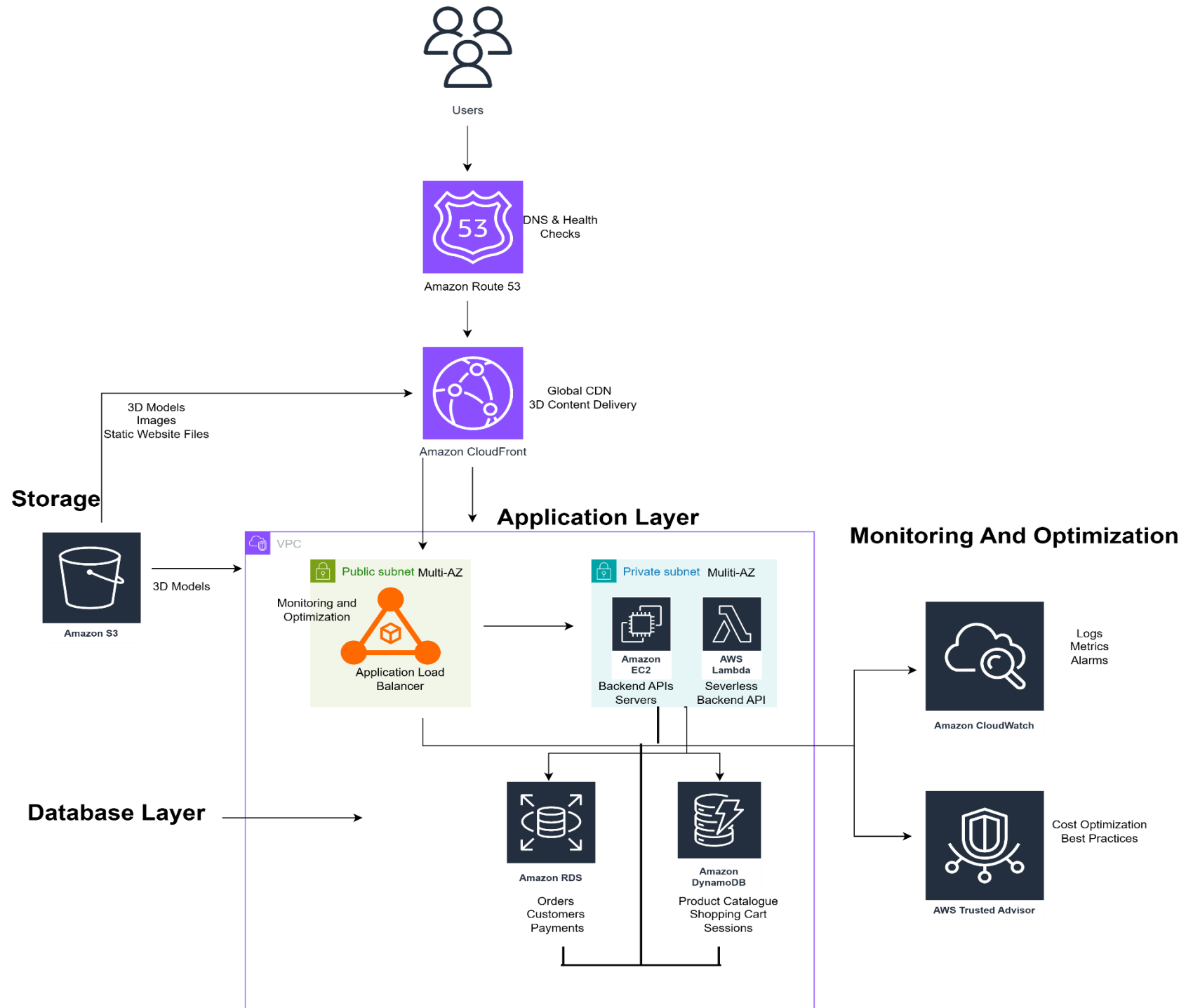




Why Each AWS Service Was Chosen

Amazon Route 53

Route 53 is used for DNS management and domain routing to ensure high availability by performing health checks and routing traffic to healthy endpoints.

Amazon CloudFront

CloudFront is used as a Content Delivery Network (CDN) to reduce latency. Since this is a 3D e-commerce platform with heavy 3D models, CloudFront ensures fast global content delivery and improved user experience.

Amazon S3

Amazon S3 is used to store 3D assets, images, and static website content. It is highly durable and integrates seamlessly with CloudFront.

Application Load Balancer (ALB)

The Application Load Balancer distributes traffic across multiple EC2 instances to ensure fault tolerance and high availability.

EC2 with Auto Scaling/ AWS Lambda

EC2 provides flexible backend compute power. Auto Scaling ensures that instances increase during traffic spikes and decrease during low demand to optimize cost. AWS Lambda can be used for a serverless backend to reduce operational overhead.

Amazon RDS (Multi-AZ)

RDS stores structured data such as customer accounts and orders. Multi-AZ deployment ensures automatic failover and high availability.

Amazon DynamoDB

DynamoDB is used for high-speed, scalable NoSQL storage such as product catalogues and user sessions. It automatically scales based on demand.

Amazon CloudWatch

CloudWatch monitors system performance, application logs, and triggers alarms when thresholds are exceeded.

AWS Trusted Advisor

Trusted Advisor helps optimize cost, security, and performance by providing AWS best practice recommendations.

2.How the Architecture Meets the 5 Requirements

High Availability

- Multi-AZ deployment for EC2 and RDS
- Load Balancer distributes traffic
- Auto Scaling replaces unhealthy instances
- Route 53 health checks
- CloudFront global edge locations
- Platform is available 24/7 with automatic failover.

Scalability

- EC2 Auto Scaling handles traffic spikes
- DynamoDB auto-scales automatically
- CloudFront reduces backend load
- Lambda (if used) scales automatically
- System handles millions of global users.

Performance

- CloudFront reduces latency
- S3 delivers 3D models efficiently
- DynamoDB provides millisecond response times
- ALB ensures optimized traffic routing
- Fast page loads and smooth 3D rendering.

Security

- VPC isolates backend resources
- Private subnets for database
- IAM roles for access control
- Secure, industry-standard cloud architecture.

Cost Optimization

- Auto Scaling prevents over-provisioning
- Lambda reduces server costs (if used)
- CloudFront reduces backend traffic
- S3 lifecycle policies for storage cost control
- Trusted Advisor recommendations
- Efficient cost management while maintaining performance.

3. Trade-Offs & Challenges

- EC2 provides control but requires management; Lambda reduces management but may have cold starts.
- RDS provides relational integrity but is more expensive than DynamoDB.
- High availability setup increases infrastructure cost.
- 3D assets require large storage and bandwidth.
- The design balances performance, availability, and cost.